Ex: NO: 7

Develop a simple calculator using XMLRPC

Aim:-

To create a python based FTP servers using the XML-RPC protocol in python that performance basic arithmetic operations like addition, subtraction, multiplication and division through client-server communication.

Description:-

XML-RPC is a protocol that uses XML to encode requests and responses and HTTP as a transport mechanism it enables communication between a client and a server where the client and a server where the client, and a server where the client invoke method hosted on the server remotely.

* The server host, arithmetic function.
* The client connects to the server to perform operation like addition, subtraction, multiplication and division.

Procedure:

* open a python IDE or terminal
* create two Separate files server.py and client.py
* write the provided server side code in server.py and client-side code in client.py.
* Run the server.py first to start the XML-RPC server.
* Run the client.py file to perform arithmetic operation remotely.
* observe the results displayed on the screen and verify the server output.

Software Requirement

* operating System: windows / Linux
* python version 3.x or above
* Modules Required: xmlrpc.server, xmlrpc.client
* Text editor or IDE - IDLE / VS code / pycharm.

Code :-

Server side code :-

```
from xmlrpc.server import SimpleXMLRPCServer
class calculator:
    def add(self, x, y):
        return x+y
    def subtract(self, x, y):
        return x-y
    def multiply(self, x, y):
        return x*y
    def divide(self, x, y):
        if y!=0:
            return x/y
        else:
            return "Error : Division by zero is not allowed"
server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Server is running on port 8000")
server.register_instance(Calculator())
server.serve_forever()
```

Client side code:

```
import xmlrpc.client

proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")

print("Addition: 10+5=", proxy.add(10,5))
print("Subtraction: 10-5=", proxy.subtract(10,5))
print("Multiplication: 10*5=", proxy.multiply(10,5))
print("Division: 10/5=", proxy.divide(10,5))
print("Division by zero: 10/0=", proxy.(10,0))
```

Output:

XML-RPC server is running on port 8000....

Client side output:

Addition: 10+5= 15

Subtraction: 10-5= 5

Multiplication: 10*5= 50

Division: 10/5= 2.0

Division by zero: 10/0= error: Division by zero is not allowed.

Result:-

The XML-RPC based calculator was successfully implemented the client communicated with server using XML-RPC performed the arithmetic operation remotely, and display accurate results. This demonstrate the used of XML-RPC protocol for remote function execution in Python.